

Assignment: Software Testing & Test Case Design

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Introduction

Software testing is a process used to verify that a system works correctly and meets user requirements. It ensures that the login system (using ID and password) functions properly, securely, and without errors.

Test Design Techniques

a) Black Box Testing

Tests functionality without knowing internal code.

Example: Checking login with valid/invalid credentials.

b) White Box Testing

Tests internal logic and code structure.

Example: Checking password validation logic.

c) Boundary Value Analysis (BVA)

Tests minimum and maximum input values.

Example: Password length (min 6, max 12 characters).

d) Equivalence Partitioning

Divides inputs into valid and invalid groups.

Example: Valid ID vs Invalid ID.

e) Decision Table Testing

Uses combinations of conditions and results.

Components of Testing

- **Test Plan:** Overall testing strategy
- **Test Case:** Step-by-step testing scenario
- **Test Data:** Input values (ID, Password)
- **Test Environment:** System/browser used
- **Test Execution:** Running test cases
- **Bug Report:** Reporting errors

Test Case Structure

Field	Description
Test Case ID	Unique number
Test Case Title	Name of test
Description	What is tested
Precondition	Required setup
Test Data	Input values
Steps	Actions
Expected Result	Expected outcome
Actual Result	Actual outcome
Attachment	Screenshot/log
Priority	High/Medium/Low
Security	Security check

Test Cases for Login System

Successful Login

- **Test Case ID:** TC_LOGIN_001
- **Test Case Title:** Valid Login with ID & Password
- **Description:** Verify user can login with correct ID and password
- **Precondition:** User account already created
- **Test Data:**
 - ID: user123
 - Password: 123456
- **Steps:**
 1. Open login page
 2. Enter valid ID
 3. Enter correct password
 4. Click "Login" button
- **Expected Result:**
User has been login successfully
- **Actual Result:**
User has been login successfully
- **Attachment:** Screenshot of dashboard
- **Priority:** High
- **Security:** Password must be encrypted

Invalid Password

- **Test Case ID:** TC_LOGIN_002
- **Test Case Title:** Login with Wrong Password
- **Description:** Verify system rejects incorrect password
- **Precondition:** User account exists
- **Test Data:**
 - ID: user123
 - Password: wrong123
- **Steps:**
 1. Enter valid ID
 2. Enter wrong password
 3. Click login
- **Expected Result:**
Error message: "Invalid password"

- **Actual Result:**
Error message displayed
- **Attachment:** Error screenshot
- **Priority:** High
- **Security:** Prevent brute force attack

Empty Fields

- **Test Case ID:** TC_LOGIN_003
- **Test Case Title:** Login with Empty Fields
- **Description:** Check validation for empty input
- **Precondition:** Login page open
- **Test Data:** Empty
- **Steps:**
 1. Leave ID and password blank
 2. Click login
- **Expected Result:**
Validation message shown
- **Actual Result:**
Validation message shown
- **Attachment:** Screenshot
- **Priority:** Medium
- **Security:** Input validation required

SQL Injection Attempt

- **Test Case ID:** TC_LOGIN_004
- **Test Case Title:** SQL Injection Test
- **Description:** Check system security against SQL injection
- **Test Data:**
 - ID: admin' OR '1'='1
 - Password: 123
- **Steps:**
 1. Enter malicious input
 2. Click login

- **Expected Result:**
Login should fail
- **Actual Result:**
Login denied
- **Attachment:** Log file
- **Priority:** High
- **Security:** Prevent SQL injection

Security Considerations

- Password encryption
- Input validation
- Protection against SQL Injection
- Account lock after multiple attempts
- Session timeout

Conclusion

This assignment shows how test design techniques and structured test cases help ensure a secure and functional login system. Proper testing improves system reliability and protects user data.